

Applied Mechanics II (Dynamics)

Course Code: BCE122
Year: I
Program: Civil Engineering

Credit Hours:3
Semester: II
Level: Bachelor

Teaching Schedule Hours/ Week			Examination Scheme				Total Marks	Remarks
			Final		Internal Assessments			
Lecture	Practical	Tutorial	Theory Marks	Practical Marks	Theory Marks	Practical Marks		
3	-	2	60	-	40	-	100	

Course Objective: To provide students with the knowledge to analyze the motion of particles and rigid bodies under the action of forces and understand the basic principles of dynamics, vibrations, and energy methods, enabling applications in civil engineering problems.

1. Introduction to Dynamics (3 hours)

- 1.1 Definition, importance , scope and Applications of Dynamics in Civil engineering
- 1.2 Branches of dynamics: Kinematics and Kinetics
- 1.3 Concept of Statics and Dynamic and their differences
- 1.4 Concept of Applied mechanics Materials (Particles, System of Particles, Deformed body, Rigid Body)
- 1.5 Fundamental Concept and principles of Newtonian Mechanics

2 Kinematics of Particles and Rigid Body (9 hours)

2.1 Rectilinear motion of particles

- 2.1.1 Position, velocity, and acceleration
- 2.1.2 Determination of motion of a particle
- 2.1.3 Uniform rectilinear and accelerated rectilinear motion
- 2.1.4 Graphical solution of rectilinear- motion problem

2.2 Curvilinear motion of particles

- 2.2.1 Position vector, velocity and acceleration
- 2.2.2 Derivatives of vector functions
- 2.2.3 Rectangular component of velocity and acceleration, Tangential and normal components, and Radial and transverse components

2.3 Rigid Body Motion

- 2.3.1 Translation and rotation of Rigid Body
- 2.3.2 General plane motion
- 2.3.3 Absolute and relative velocity in plane motion
- 2.3.4 Instantaneous center of rotation
- 2.3.5 Motion about a fixed point
- 2.3.6 Three-dimensional motion of a particle relative to a rotating frame; Coriolis acceleration

3 . Kinetics of Particles (12 hours)

3.1 Newton's Second Law of Motion

- 3.1.1 Introduction
- 3.1.2 Equation of motion , Dynamic equilibrium, and Alembert's Principles
- 3.1.3 Linear and angular momentum of a particle
- 3.1.4 Rate of change of linear and angular momentum
- 3.1.5 Rectangular, Radial, and Transverse and Normal and Tangential component of motion
- 3.1.6 Conservation of Momentum and Newton's Law of Gravitation

3.2 Energy and Momentum Methods

- 3.2.1 Work done by a force, Spring and Gravity
- 3.2.2 Potential , kinetic and Spring energy of particles
- 3.2.3 Principles of work and energy: applications
- 3.2.4 Power and efficiency
- 3.2.5 Conservation of energy
- 3.2.6 Principle of impulse and momentum
- 3.2.7 Impulsive motion and impact
- 3.2.8 Direct central and oblique impact

4 . System of Particles (5 hours)

- 4.1 System of particles and Newton's Second Law of Motion
- 4.2 Linear and angular moment for a system of particles
- 4.3 Motion of the mass centre, Equation of motion and Dynamic Equilibrium
- 4.4 Conservation of momentum
- 4.5 Kinetic and potential energy of system of particles
- 4.6 Work energy principles; Conservation of energy for a system of particles
- 4.7 Principles of impulse and momentum for a system of particles
- 4.8 Steady stream of particles
- 4.9 System with variable mass

5 Kinetics of Rigid Bodies (12 hours)

5.1 Newton's Second law of motion

- 5.1.1 Equation of motion for a rigid body in plane motion
- 5.1.2 Angular momentum of a rigid body in plane motion
- 5.1.3 Plane motion of rigid body: D'Alembert's principle
- 5.1.4 Application of rigid body motion in the plane
- 5.1.5 System of rigid bodies
- 5.1.6 Constrained motion in the plane

5.2 Energy and Momentum Methods

- 5.2.1 Principle of work and energy for a rigid body
- 5.2.2 Work done by external forces
- 5.2.3 Kinetic energy for a system
- 5.2.4 Conservative and non-conservative systems
- 5.2.5 Work – energy applications
- 5.2.6 Impulse and momentum for systems for rigid bodies
- 5.2.7 Conservation of angular and linear momentum
- 5.2.8 Impulsive motion and eccentric impact

6 Mechanical Vibration (4 hours)

- 6.1 Introduction to Mechanical Vibration, Vibration types and Sources
- 6.2 Application of Mechanical Vibration in Civil Engineering
- 6.3 Simple Harmonic Motion
- 6.4 Undamped and Damped Free and Forced Vibration

Tutorials (15 hours)

Unit-specific exercises focused on the application of dynamics in civil engineering.

Reference books:

1. F.P. Beer, and E.R. Johnson, "Vector Mechanics for Engineers", Tata McGraw Hill Publishing Co. Ltd.
2. I. H. Shames, "Engineering Mechanics – Statics and Dynamics", New Delhi, Prentice Hall of India.
3. P. P. Egor, "Engineering Mechanics of Solids", New Delhi, Prentice Hall of India.
4. R. Hibbeler, "Engineering Mechanics: Statics & Dynamics", Pearson
5. Bedford and W. Fowler, "Engineering Mechanics: Statics & Dynamics". Pearso

Evaluation Scheme:

The questions will cover all the chapters of the syllabus. The evaluation scheme will be as indicated in the table below:

Chapters	Hours	Marks Distribution*
1	3	4
2	9	12
3	12	16
4	5	7
5	12	16
6	4	5
Total	45	60

*There may be minor deviation in marks distribution

